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Pil E.A.

Academician RANH, Professor, Dr. Sc. Eng., St. Petersburg

THE EFFECT OF THE VARIABLE X4 ON 3D GRAPHS FOR VEL WITH NEGATIVE VALUES OF VARIABLES

Two years ago, the Nobel Prize in Economics was awarded to three American economists who significantly improved the understanding of the role of banks in the economy, especially during financial crises, by giving money to banks and the population to restore purchasing activity in the country. Here it is worth recalling at once that in a number of countries, such as Sweden, Switzerland, Denmark and Japan, banks issued some customers with a negative rate, i.e. customers returned an amount less than they received [1, 2, 3].

This article considers the issue of calculating the values of the lower critical volume of the Vel spherical economic shell and constructing 3D drawings when the values of three variables change, i.e. $Vel (GDP_{el}) = f(X1, X2, X3, X4, X5)$. In some cases, when the Vel values had small minima and maxima compared to the last calculated Vel value, then 3D drawings were built from the calculation so that the reader would understand more clearly how these changes occur.

Here, Figure 1 shows a 3D Vel graph when the values of the variables were as follows $X1 = X3 = -0.1 \dots -1$, $X2 = X5 = -1 \dots -0.1$, $X4 = 0.76 \dots 0.1$. Here the Vel decreases significantly by a factor of 1000. In the following Figure 2 the depicted 3D Vel graph at variables $X1 = X3 = X5 = -1 \dots -0.1$, $X2 = -0.1 \dots -1$, $X4 = 0.71 \dots 0.29$ decreases by 246.37 times.

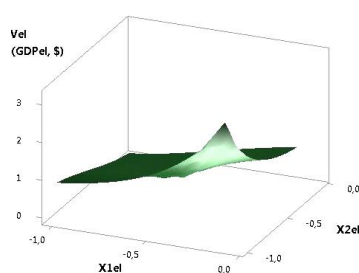


Fig. 1. 3D of Vel (GDPel)

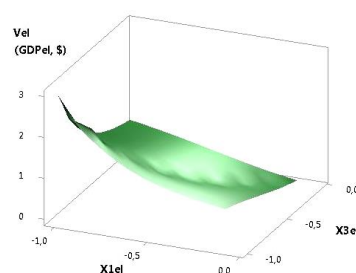


Fig. 2. 3D of Vel (GDPel)

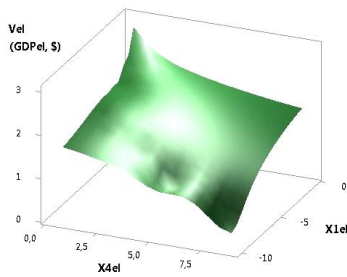


Fig. 3. 3D of Vel (GDPel)

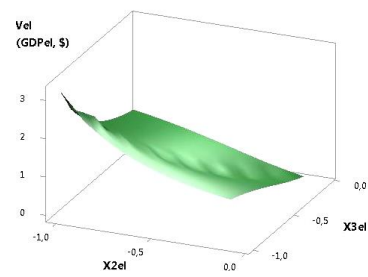


Fig. 4. 3D of Vel (GDPel)

The following two figures 3 and 4 show 3D Vel graphs when the variables were $X1 = -10 \dots -1$, $X2 = -10 \dots -1$, $X3 = -10 \dots -1$, $X4 = 0.71 \dots 8.8$, $X5 = -1 \dots -0.1$ and $X1 = -1 \dots -10$, $X2 = -10 \dots -1$, $X3 = -10 \dots -1$, $X4 = 0.76 \dots 0.29$. As we can see, the plotted 3D Vel graph in Fig. 3 has a maximum of 2.02 at point 5, and in Fig. 4. it will decrease by 264.58 times.

The plotted 3D Vel graph in Figure 5 for variables $X1 = -10 \dots -1$, $X2 = -10 \dots -10$, $X4 = 1.13 \dots 0.88$, $X5 = -1 \dots -0.1$ decreases by 127.85 times. From the following Figure 6 it can be seen that for variables $X1 = -10 \dots -1$, $X2 = -10 \dots -1$, $X3 = -1 \dots -10$, $X4 = 0.883 \dots 0.11$, $X5 = -1 \dots -0.1$ Vel decreases very significantly by 7821.54 times.

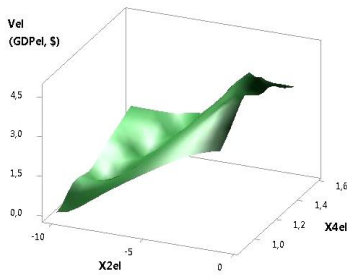


Fig. 5. 3D of Vel (GDPel)

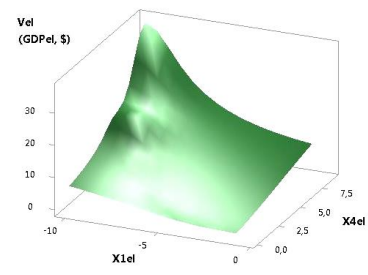


Fig. 6. 3D of Vel (GDPel)

Figures 7 and 8 were plotted at $X1 = -10 \dots -1$, $X2 = -10 \dots -1$, $X4 = 2.87 \dots 0.29$, $X5 = -1 \dots -0.1$ and $X1 = -1 \dots -10$, $X2 = -10 \dots -1$, $X4 = 1.13 \dots 0.29$, $X5 = -1 \dots -0.1$. Here in Figures 7 and 8, the Vel values decrease by 1000 times and 393.12 times, respectively.

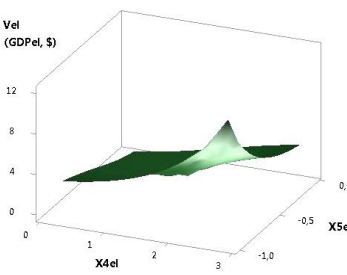


Fig. 7. 3D of Vel (GDPel)

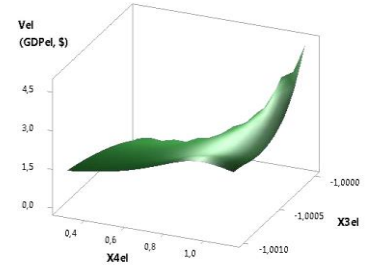


Fig. 8. 3D of Vel (GDPel)

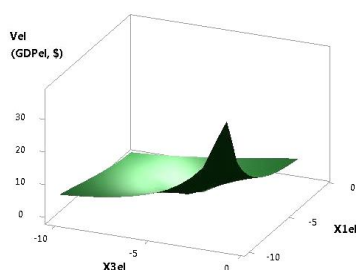


Fig. 9. 3D of Vel (GDPel)

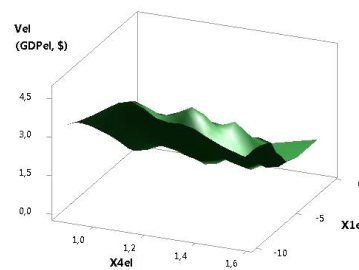


Fig. 10. 3D of Vel (GDPel)

The following Figure 9 was plotted at $X1 = X2 = -1 \dots -10$, $X3 = -10 \dots -1$, $X4 = 0.11 \dots 0.83$, $X5 = -0.1 \dots -1$, where Val increases significantly by 7821.54 times. Figure 10 plotted at $X1 = -1 \dots -10$, $X2 = X3 = -10 \dots -1$, $X4 = 0.8 \dots 1.13$, $X5 = -0.1 \dots -1$ shows that Val values increase by 127.85 times.

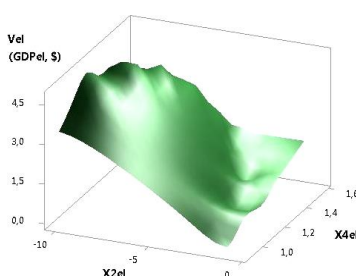


Fig. 11. 3D of Vel (GDPel)

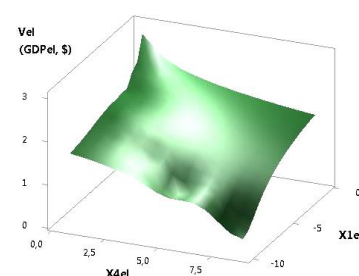


Fig. 12. 3D of Vel (GDPel)

The last figures 11 and 12 were plotted at $X1 = -10 \dots -1$, $X2 = X3 = -1 \dots -10$, $X4 = 0.88 \dots 1.13$, $X5 = -0.1 \dots -1$ and $X1 = X2 = -10 \dots -1$, $X3 = -1 \dots -10$, $X4 = 8.8 \dots 0.71$, $X5 = -0.1 \dots -1$. In these examples in Figure 11, the Vel variables increase by a factor of 127.85, and in Figure 12 they have a minimum of 2 at point 7.

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